

Project Proposal

Campus Health Appointment & Medicine Management System

Members (BAD 542)

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1. Project Domain

The **Campus Health Appointment & Medicine Management System** is a web-based application that enables students to book appointments with university doctors through a secure online platform. Users authenticate using the University's Microsoft Active Directory account. Doctors manage appointments and prescriptions, while administrators oversee users, doctors, and appointment schedules. The system integrates with the **Campus Emergency & Safety Alert System** (SL systems) to retrieve active campus emergency alerts before confirming appointments and to share appointment schedules when required during campus emergencies. The application will be deployed on a Linux VPS using Docker Compose, Nginx Reverse Proxy, Let's Encrypt SSL, Prisma ORM, Azure Key Vault, and integrates external AI and peer APIs.

2. System Architecture

Users
|
Nginx Reverse Proxy (HTTPS)
|
React Frontend
|
Node.js Express API

```
|-- Prisma ORM -> MySQL
|-- Microsoft Azure AD
|-- OpenAI API
|-- Azure Key Vault
|-- Peer Team API
```

3. Database Schema (ERD)

User

id(PK), name, email, role, adId

Doctor

id(PK), name, specialty, room

Appointment

id(PK), studentId(FK), doctorId(FK), appointmentDate, symptoms, status

Prescription

id(PK), appointmentId(FK), medicine, dosage, instruction

Relationships:

User (1) ---- (*) Appointment

Doctor (1) ---- (*) Appointment

Appointment (1) ---- (1) Prescription

4. RBAC

Role	Permissions
Student	Book/cancel appointments, view prescriptions
Doctor	Manage appointments, create prescriptions
Administrator	Manage users, doctors, appointments, configuration

5. Third-Party API

DeepSeek API

Purpose: Generate concise summaries of student symptoms to assist doctors before appointments.

6. Peer API Integration

Consume API

GET /api/alerts with x-api-key

Purpose:

Retrieve active campus emergency alerts from the Campus Emergency & Safety Alert System. Before confirming an appointment, the system checks for any active emergency alerts. If an emergency affects the campus or clinic, students are informed and new appointments may be temporarily unavailable.

Expose API

GET /api/appointments?date={date} with x-api-key

Purpose:

Allow the Campus Emergency & Safety Alert System to retrieve appointment schedules for a selected date. During a campus emergency, the system can identify students who have appointments, send emergency notifications, and assist with appointment rescheduling if necessary.

7. Technology Stack

Frontend: React.js

Backend: Node.js + Express

Database: MySQL

ORM: Prisma

Authentication: Microsoft Azure AD (MSAL/OIDC)

Authorization: JWT + RBAC

Secrets: Azure Key Vault

Reverse Proxy: Nginx

SSL: Let's Encrypt

Deployment: Docker Compose
Cloud: Azure Cloud Linux VPS
Version Control: GitHub

8. Expected Outcomes

A secure university health management system that enables students to book medical appointments online, allows doctors to manage appointments and prescriptions, and provides administrators with centralized system management. The system integrates Microsoft Azure Active Directory for authentication, OpenAI for symptom summarization, and peer APIs with the **Campus Emergency & Safety Alert System(SL System)** to improve coordination during campus emergencies. The application is securely deployed on an Azure Linux VPS using Docker Compose, Prisma ORM, Azure Key Vault, Nginx Reverse Proxy, and Let's Encrypt SSL.